

逐步构建过程

$$\begin{aligned}
& X_s = XQ_s \\
& \min_{Q_s} \sum_{s=1}^S \|Q_s\|_F^2 \\
& \text{s.t. } X_s = XQ_s + E_{1,s} \\
& \min_{P,Z,Q,E} \sum_{s=1}^S \|Z_s\|_* + \beta \sum_{s=1}^S \|Q_s\|_F^2 + \gamma \sum_{s=1}^S \|E_s\|_{2,1} + \delta \sum_{s=1}^S \|X_s - XQ_s\|_F^2 \\
& \quad + \kappa \sum_{s=1}^S \|P_s X_s - P_s X_s Z_s\|_F^2 + \lambda \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\
& \text{s.t. } X_s = X_s Z_s + E_s, P_s^T P_s = I
\end{aligned}$$

其中：

$X = [X_1, X_2, \dots, X_S] \in \mathbb{R}^{D \times N}$, 是多模态数据, D 为原始数据的特征数, N 为所有模态数据的样本总数 $N = \sum^S n_s$

$X_s \in \mathbb{R}^{D \times n_s}$, 是每个模态的数据

$P_s \in \mathbb{R}^{d \times D}$, d 为降维之后特征数

$Q_s \in \mathbb{R}^{N \times n_s}$, 是筛选矩阵, 将单模态数据从原始数据中提取出来, n_s 为单个模态数据的样本数。

$Z_s \in \mathbb{R}^{n_s \times n_s}$

$E_s \in \mathbb{R}^{D \times n_s}$

引入辅助变量 J

$$\begin{aligned}
& \min_{P,Z,Q,E,J} \sum_{s=1}^S \|J_s\|_* + \beta \sum_{s=1}^S \|Q_s\|_F^2 + \gamma \sum_{s=1}^S \|E_s\|_{2,1} \\
& + \delta \sum_{s=1}^S \|X_s - XQ_s\|_F^2 + \kappa \sum_{s=1}^S \|P_s X_s - P_s X_s Z_s\|_F^2 \\
& \quad + \lambda \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\
& \text{s.t. } X_s = X_s Z_s + E_s, P_s^T P_s = I, Z = J
\end{aligned}$$

⇓ 构建增广拉格朗日函数

$$\begin{aligned}
& \mathcal{L}(Z, P, Q, E, J, \Gamma, \Delta) \sum_{s=1}^S \|J_s\|_* + \beta \sum_{s=1}^S \|Q_s\|_F^2 + \gamma \sum_{s=1}^S \|E_s\|_{2,1} \\
& + \delta \sum_{s=1}^S \|X_s - XQ_s\|_F^2 + \kappa \sum_{s=1}^S \|P_s X_s - P_s X_s Z_s\|_F^2 \\
& \quad + \lambda \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\
& + \frac{\mu}{2} \sum_{s=1}^S \left(\left\| X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right\|_F^2 + \left\| Z_s - J_s + \frac{\Delta}{\mu} \right\|_F^2 \right)
\end{aligned}$$

一次固定一个模式

$$\begin{aligned}
& \mathcal{L}(Z, P, Q, E, J, \Gamma, \Delta) \|J_s\|_* + \beta \|Q_s\|_F^2 + \gamma \|E_s\|_{2,1} \\
& + \delta \|X_s - XQ_s\|_F^2 + \kappa \|P_s X_s - P_s X_s Z_s\|_F^2 \\
& \quad + \lambda \sum_{t=1, t \neq s}^S \text{HSIC}(Z_s, Z_t) \\
& + \frac{\mu}{2} \left(\left\| X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right\|_F^2 + \left\| Z_s - J_s + \frac{\Delta}{\mu} \right\|_F^2 \right)
\end{aligned}$$

HSIC约束选择内积核函数 $K_s = Z_s^T Z_s$

$$\sum_{t=1,t\neq s}^S \text{HSIC}(Z_s, Z_t) = \sum_{t=1,t\neq s}^S \text{Tr}(H K_s H K_t) = \sum_{t=1,t\neq s}^S \text{Tr}(Z_s H K_t H Z_s^T) = \text{Tr}(Z_s K Z_s^T)$$

其中：

$$K = \sum_{t=1,t\neq s}^S H K_t H$$

$$\begin{aligned} \mathcal{L}(Z, P, Q, E, J, \Gamma, \Delta) & \|J_s\|_* + \beta \|Q_s\|_F^2 + \gamma \|E_s\|_{2,1} \\ & + \delta \|X_s - X Q_s\|_F^2 + \kappa \|P_s X_s - P_s X_s Z_s\|_F^2 \\ & + \lambda \text{Tr}(Z_s K Z_s^T) \\ & + \frac{\mu}{2} \left(\left\| X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right\|_F^2 + \left\| Z_s - J_s + \frac{\Delta}{\mu} \right\|_F^2 \right) \end{aligned}$$

其中：

$$K = \sum_{t=1,t\neq s}^S H K_t H$$

$H = I - \frac{1}{n} \mathbf{1} \mathbf{1}^T$ 是中心化矩阵,

⇓ 更新

通过解析解更新

step 1 J

$$J_{k+1} = \Theta(\frac{1}{\mu}) \left(Z + \frac{\Delta}{\mu} \right)$$

step 2 Z

$$\min_{Z_s} \kappa \|P_s X_s - P_s X_s Z_s\|_F^2 + \lambda \text{Tr}(Z_s K Z_s^T) + \frac{\mu}{2} \left(\left\| X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right\|_F^2 + \left\| Z_s - J_s + \frac{\Delta}{\mu} \right\|_F^2 \right)$$

范数转化成tr的形式

$$\begin{aligned} \min_{Z_s} \kappa & \text{Tr}((P_s X_s - P_s X_s Z_s)(P_s X_s - P_s X_s Z_s)^T) + \lambda \text{Tr}(Z_s K Z_s^T) \\ & + \frac{\mu}{2} \left(\text{Tr} \left(\left(X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right) \left(X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right)^T \right) + \text{Tr} \left(\left(Z_s - J_s + \frac{\Delta}{\mu} \right) \left(Z_s - J_s + \frac{\Delta}{\mu} \right)^T \right) \right) \end{aligned}$$

求导

function:

$$\begin{aligned} f = & \kappa \cdot \text{tr}((P \cdot X - P \cdot X \cdot Z) \cdot (P \cdot X - P \cdot X \cdot Z)^\top) + \lambda \text{tr}(Z \cdot K \cdot Z^\top) \\ & + \mu/2 \cdot (\text{tr}((X - X \cdot Z - E + \text{Lambda}/2) \cdot (X - X \cdot Z - E + \text{Lambda}/2)^\top) + \text{tr}((Z - J + \text{Delta}/2) \cdot (Z - J + \text{Delta}/2)^\top)) \end{aligned}$$

gradient:

$$\begin{aligned} \frac{\partial f}{\partial Z} &= \textit{lambda} \cdot Z \cdot K^\top - 2 \cdot \textit{kappa} \cdot (P \cdot X)^\top \cdot (P \cdot X - P \cdot X \cdot Z) \\ &+ \textit{lambda} \cdot Z \cdot K - \textit{mu} \cdot X^\top \cdot (X - X \cdot Z - E + 1/2 \cdot \textit{Lambda}) + \textit{mu} \cdot (Z - J + 1/2 \cdot \textit{Delta}) \\ &= 0 \end{aligned}$$

→

$$(2\kappa A^T A + \mu X^T X + \mu I)Z + Z\lambda(K^T + K) = 2\kappa A^T A + \mu X^T X - \mu X^T(E - \Lambda/2) + \mu J - \mu/2\Delta$$

其中:

$$A = PX, K = \sum_{t=1, t \neq s}^S H K_t H, K_t = Z_t^T Z_t$$

$H = I - \frac{1}{n}11^T$ 是中心化矩阵,

$$\mathbf{Z} = \text{sylvester}(\mathbf{A}, \mathbf{B}, \mathbf{C});$$

step 3 P

$$\begin{aligned} &\min_{P_s} \kappa \|P_s X_s - P_s X_s Z_s\|_F^2 \\ &= \min_{P_s} \kappa \text{Tr}(PX(Z - I)(Z - I)^T X^T P^T) \end{aligned}$$

转化为求特征值

$$X(\kappa(Z - I)(Z - I))X^T p = \xi p$$

选前lower_dim个向量拼接成 P_s^T

step 4 Q

$$\min_{Q_s} \beta \|Q_s\|_F^2 + \delta \|X_s - XQ_s\|_F^2$$

求导=0

$$Q_s = (\beta I + \delta X^T X)^{-1} \delta X^T X_s$$

也可以换成梯度下降法求解

$$Q_{s,k+1} = Q_{s,k} - \eta(2\beta Q_{s,k} - 2\delta X^T(X_s - XQ_{s,k}))$$

step 5 E

$$\min_{E_s} \gamma \|E_s\|_{2,1} + \frac{\mu}{2} \left\| X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right\|_F^2$$

$$V = X_s - X_s Z_s + \frac{\Gamma}{\mu}$$

$$[E^*]_{:,i} = \begin{cases} [V_{:,i}]_2 - \alpha & \text{if } \|V_{:,i}\|_2 > \alpha \\ 0 & \text{otherwise} \end{cases}$$

where $[E^*]_{:,i}$ is the column of the optimal solution E^* .

step 5 others

$$\begin{aligned}\Gamma_{k+1} &= \Gamma_k + \mu_k (X - XZ_{k+1} - E_{k+1}) \\ \Delta_{k+1} &= \Delta_k + \mu_k (Z_{k+1} - J_{k+1}) \\ \mu_{k+1} &= \min(\rho\mu_k, \mu_{\max})\end{aligned}$$